

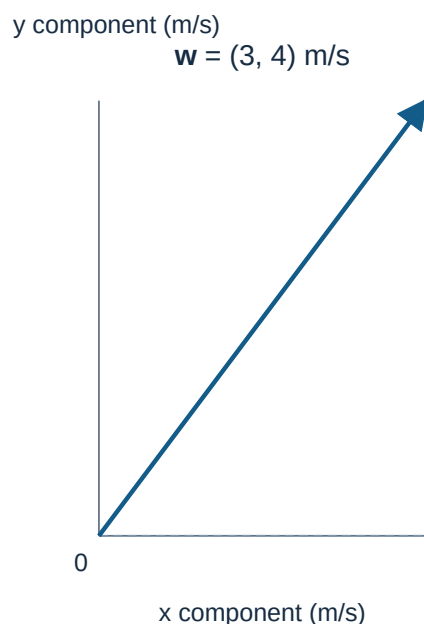
Motion in two dimensions: vector representation and subtraction; relative velocity in a plane · CORE1A

Vector representation and subtraction

Intrinsic concept difficulty: MEDIUM

Before any number is subtracted or added, name the frame. Declare which direction along each axis counts as positive -- here, east is positive x and north is positive y -- because a signed component only means something once that choice is fixed.

A vector w with components $(3, 4)$ m/s is read directly off those axes: 3 m/s east and 4 m/s north. The vector and its magnitude are two different descriptions of the same physical quantity. The magnitude (speed) is obtained by treating the two perpendicular components as the legs of a right triangle: $\text{speed} = \sqrt{3^2 + 4^2} = \sqrt{25} = 5$ m/s. Squaring removes the sign from each component, so the magnitude is always taken nonnegative -- but the original signed pair $(3, 4)$ is still required afterwards to say which way the vector points. Stating '5 m/s' alone never answers a direction question.



Vector w , drawn from the origin with its signed components. Frame: common east/north Cartesian frame; east positive x , north positive y . Quantitative diagram.

Subtraction of two vectors is defined as addition of the reversed second vector: $P - Q = P + (-Q)$. Reversing Q keeps its length exactly the same and flips its direction end for end; reversal also flips the sign of every one of its signed

components.

Take $P = (6, 0)$ m/s and $Q = (0, 8)$ m/s in the common east/north frame. Reversing Q gives $-Q = (0, -8)$ m/s. As a free vector, $-Q$ may be translated to the head of P without being rotated. Adding tail-to-head, the resultant is $P + (-Q) = (6 + 0, 0 + (-8)) = (6, -8)$ m/s. This graphical route must agree with the direct component-by-component subtraction, $(P_x - Q_x, P_y - Q_y) = (6 - 0, 0 - 8) = (6, -8)$ m/s -- and it does. Applying the right-triangle bridge, the magnitude is $\sqrt{6^2 + 8^2} = \sqrt{100} = 10$ m/s.

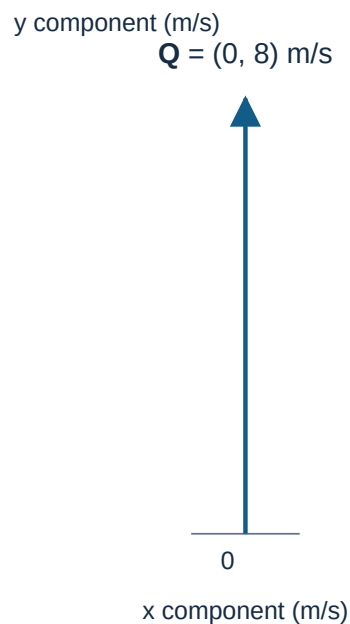
$$P - Q = P + (-Q)$$

Vector subtraction is defined as addition of the reversed second vector.

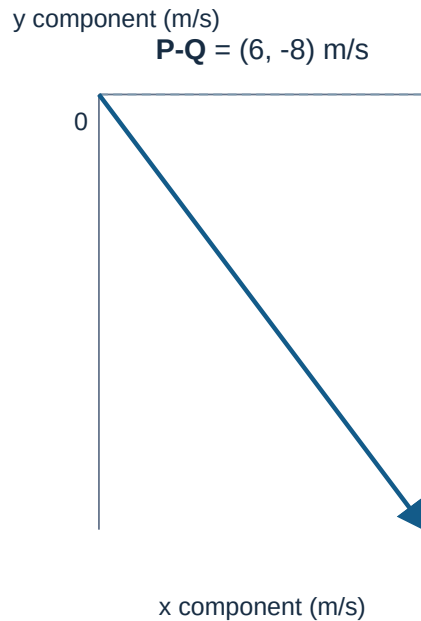
1. P : the first vector in the declared common frame.
2. Q : the second vector, expressed in the same frame.
3. $-Q$: Q reversed -- identical magnitude, opposite direction.

Applies when:

1. P and Q are expressed along the same declared axes.
2. Free vectors may be translated (not rotated) for the tail-to-head construction.



Q as given. This is the vector that gets reversed in the construction. Frame: common east/north Cartesian frame; east positive x , north positive y . Quantitative diagram.



The resultant P-Q, reached by adding P and the reversed -Q tail-to-head; it agrees with the direct component subtraction. Frame: common east/north Cartesian frame; east positive x, north positive y. Quantitative diagram.

Relative velocity in a plane

Intrinsic concept difficulty: HARD

'A relative to B' is built from same-time positions, not memorised as a bare formula. If r_A and r_B are the positions of A and B at the same instant in one common frame, then following the path origin $\rightarrow B \rightarrow A$ gives $r_A = r_B + r_{A/B}$, so $r_{A/B} = r_A - r_B$: the position of A measured from B.

Writing this relation at two instants t_1 and t_2 and subtracting gives $\Delta r_{A/B} = \Delta r_A - \Delta r_B$. Dividing every term by the same nonzero time interval turns each side into an average velocity; for the constant velocities used in this teaching case, $v_{A/B} = v_A - v_B$. Both displacement changes must be divided by the same shared interval -- comparing changes measured over different intervals would not describe how one simultaneous separation evolves.

Apply this to A moving at (6, 0) m/s and B moving at (0, 8) m/s in a common east/north frame: $v_{A/B} = (6, 0) - (0, 8) = (6, -8)$ m/s. By the right-triangle bridge, the magnitude is $\sqrt{6^2 + 8^2} = \sqrt{100} = 10$ m/s. The eastward component is positive and the northward component is negative, so the vector points southeast.

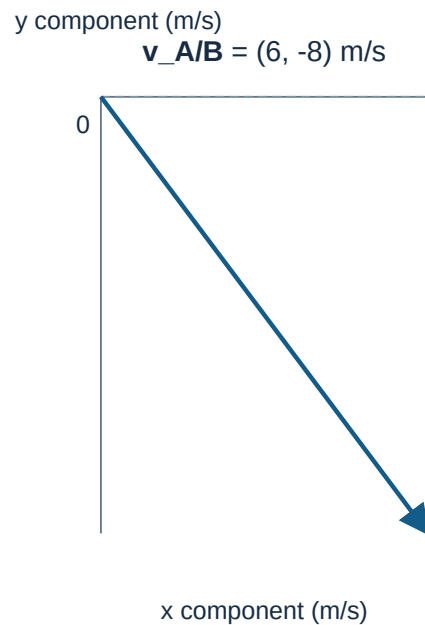
$$v_{A/B} = v_A - v_B$$

For constant velocities, the relative velocity of A with respect to B is the vector difference of their common-frame velocities.

1. v_A : velocity of A in the common frame.
2. v_B : velocity of B in the same frame, same instant.
3. $v_{A/B}$: velocity of A relative to B, i.e. A as measured by an observer moving with B.

Applies when:

1. Use the same observation times and a common time interval.
2. Express all vectors along parallel, nonrotating Cartesian axes.
3. Use a classical (non-relativistic) kinematic model.



The resultant relative-velocity vector A relative to B, reconciled with the component calculation above. Frame: common east/north Cartesian frame; east positive x, north positive y. Quantitative diagram.

Swapping the observer reverses the vector: $\mathbf{v}_{B/A} = -(\mathbf{v}_{A/B})$. Since A relative to B is (6, -8) m/s, B relative to A is (-6, 8) m/s -- every component changes sign. The magnitude is unchanged, because the right-triangle bridge only uses squared components, and squaring erases a sign flip: $\sqrt{(-6)^2 + 8^2} = \sqrt{100} = 10 \text{ m/s}$, matching the original 10 m/s. Adding the two relative vectors together gives exactly zero, which is a standing check for this reversal relation on any pair of velocities.

Hints, answers and repair
